EPSAC Control Design for Stabilizing Cart Position and Pendulum Angle

1 Introduction

This document explains an EPSAC (Economic Predictive Self-Adaptive Control) strategy applied to a Segway-like system (a wheeled inverted pendulum). The controller combines a model predictive control (MPC) approach with an adaptive update of the system model. The goal is to stabilize both the horizontal position of the cart and the pendulum angle by optimizing future control actions under constraints.

2 EPSAC Controller Overview

The controller operates on a discretized version of the plant's linearized state-space model. It uses a finite prediction horizon and a shorter control horizon to compute an optimal sequence of control inputs. Only the first input is applied, and the process is repeated at each time step.

3 Key Tuning Parameters and Their Roles

3.1 Prediction Horizon (N_p)

- **Definition:** The number of future steps over which the system's behavior is predicted.
- **Role:** A longer N_p allows the controller to plan further ahead, potentially improving performance by anticipating future events.
- **Tuning Effects:**
  - *Longer horizon:* May improve tracking performance and robustness but increases computational complexity.
  - *Shorter horizon:* Reduces computation but may lead to myopic decisions and poorer performance in the presence of delays or disturbances.

3.2 Control Horizon (N_c)

- **Definition:** The number of control moves that are optimized within the prediction horizon.
- **Role:** Determines the degree of freedom the optimizer has to shape the control trajectory.
- **Tuning Effects:**
  - *Longer control horizon:* Can lead to more flexible control actions but also increases the size of the optimization problem.
  - *Shorter control horizon:* Simplifies optimization and reduces computational load, though it may restrict the controller's ability to react to disturbances.

3.3 Control Effort Weight (λ)

- **Definition:** A scalar weight in the cost function that penalizes the magnitude of the control input.
- **Role:** Balances the trade-off between tracking performance and minimizing control energy.
- **Tuning Effects:**
  - *Higher λ:* Places greater emphasis on reducing control effort, which may lead to smoother, but potentially slower, responses.
  - *Lower λ:* Allows for more aggressive control actions to achieve faster tracking but at the cost of increased energy consumption and potential actuator saturation.

3.4 Input Constraints ($u_{\min}, u_{\max}$)

- **Definition:** Bounds on the allowable control input, ensuring that the actuator operates within safe limits.
- **Role:** Prevents the optimizer from suggesting control inputs that are physically impossible or that might damage the system.
- **Tuning Effects:**
  - *Tighter constraints:* Improve safety and reliability, but may limit performance if the required control exceeds these bounds.
  - *Looser constraints:* Allow for more aggressive control but risk actuator saturation or hardware damage.

3.5 Adaptive Model Update Gain

- **Definition:** A small adaptation gain (e.g., 0.001) used to update the estimated system matrices A_{est} and B_{est}.
- **Role:** Adjusts the model used for prediction based on the error between predicted and actual states.
- **Tuning Effects:**
  - *Higher gain:* Leads to faster adaptation but may cause instability if the model updates are too aggressive.
  - *Lower gain:* Provides more gradual adaptation, which may be safer but slower to react to model discrepancies.

4 EPSAC Controller Implementation Summary

The EPSAC controller for the Segway-like system proceeds as follows:

1. **Model Discretization:** The continuous-time state-space model is discretized using a sampling time T_s (e.g., 0.1 seconds).
2. **Optimization Problem Formulation:** A quadratic cost function is defined over the prediction horizon N_p, penalizing both tracking error and control effort. The cost function incorporates the weight λ.
3. **Constraint Handling:** The control input is constrained between $u_{\min}$ and $u_{\max}$.
4. **Control Horizon Optimization:** Only N_c control moves are optimized; beyond N_c, the control action is held constant.
5. **Adaptive Model Update:** After applying the control input, the model estimates A_{est} and B_{est} are updated based on the prediction error, scaled by the adaptation gain.
6. **Receding Horizon Implementation:** The first control input from the optimized sequence is applied, and the process is repeated at the next time step.

5 Conclusion

The EPSAC approach offers a powerful method to stabilize both the cart position and the pendulum angle by:

- Predicting future system behavior over a specified horizon (N_p).
- Optimizing a sequence of control actions over a shorter horizon (N_c) while penalizing control effort (λ).

- Ensuring safety via input constraints ($u_{\min}$ and $u_{\max}$).
- Adapting the model in real time with a small adaptation gain.

Tuning these parameters directly affects the controller's responsiveness, energy usage, and robustness. Proper selection of N_p, N_c, λ, and the adaptation gain is essential for achieving optimal performance in the face of model uncertainties and external disturbances.